Lesson notes · for you

Passwords nobody can guess

Cyber security · Level 1 · Lesson 1

Time	Materials
15 min	Paper and a pen; A phone or computer, if you have one

Learning goals

- Explain how programs guess passwords.
- Make a passphrase from random words.
- Use a different password for every account.
- Keep passwords secret, even from friends.

Activity: The passphrase challenge

1. Warm-up (3 min): Write five passwords a guessing program would try first for a made-up student called Musa (his name, his birthday, his team…).
2. Pick (4 min): Open a book at four random pages and take the first noun on each, or pull four words from a hat.
3. Picture (5 min): Draw one funny scene that links your four words. The sillier, the easier to remember.
4. Test (3 min): Turn the paper over and write your four words from memory. Then explain to a partner why your phrase beats Musa's passwords.

Use practice words only: never write a real password on the sheet.

Discussion

- Which of your accounts matters most, and why?
- Why don't symbols on their own help much?
- Where could you keep many passwords safely?

Check yourself

- One account I will fix, and how.

Try it this week

Pick four random words and picture one funny scene that links them. If you still remember it at bedtime, make a new passphrase of your own for your email account — with a parent or guardian's help if you like.

Worksheet

Passwords nobody can guess

Cyber security · Level 1 · Lesson 1

Name _____ Class _____ Date _____

1 **Your best friend wants your game password to collect your daily rewards. Do you share it?**

◯ Yes, and I'll change it next month ◯ Yes — they're my best friend

◯ No — passwords stay secret, even from friends

2 **Swapping letters for symbols, like P@ssw0rd, makes a short password hard to guess.**

◯ True ◯ False

3 **Which of these would a guessing program try early? Choose all that apply.**

☐ Your pet's name ☐ Four random words you picked yourself ☐ Your favourite football team

☐ Your name and birth year ☐ 123456

4 **Complete the rule.**

A strong password is _____ , made of random words, and _____ for every account.

Word bank: short · the same · long · different

5 **Match each situation to the safest habit.**

You use a computer in a cyber café	• • • •	Change that password, and anywhere you reused it
A game website you used gets hacked	• • • •	Say no, kindly, and keep it secret
You have too many passwords to remember	• • • •	Sign out, and don't save the password
A classmate asks for your password	• • • •	A password manager, or a notebook kept at home

✎ **Which of your accounts would hurt most if someone got into it? What will you change about its password?**

Check your answers

Passwords nobody can guess

Cyber security · Level 1 · Lesson 1

1 **No — passwords stay secret, even from friends**

A shared password can be passed on, used after a fight, or typed on a phone that isn't safe — and a month is a long time for all of that to happen.

2 **False**

Guessing programs know these swaps and try them early.

3 **Your name and birth year; Your favourite football team; 123456; Your pet's name**

Names, birthdays, teams, pets and number runs are the first things programs try, and many of them are on your profile for anyone to see.

4 **A strong password is long, made of random words, and different for every account.**

Length is what slows guessing programs down, and a different password for each account means one leak can't open all the others.

5 **You use a computer in a cyber café — Sign out, and don't save the password**
A game website you used gets hacked — Change that password, and anywhere you reused it
You have too many passwords to remember — A password manager, or a notebook kept at home
A classmate asks for your password — Say no, kindly, and keep it secret

Each habit closes a different door: shared computers remember what you typed, leaked passwords travel to other sites, memory runs out, and secrets you share spread.